

Tara Finance Agent - Design Document

1. System Overview

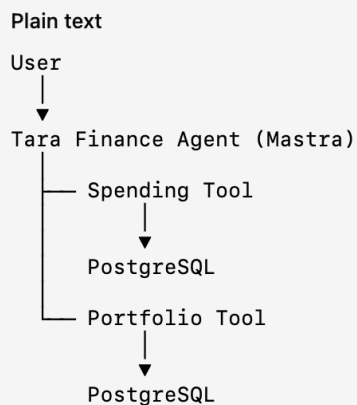
Tara Finance Agent is a financial analytics assistant built using Mastra, PostgreSQL, and TypeScript.

The system enables users to ask natural language questions related to:

- * Spending analysis
- * Category spending
- * Merchant spending
- * Spending trends
- * Spending comparisons
- * Portfolio analysis
- * Mutual fund performance
- * Investment returns

The agent converts user questions into structured tool calls and returns insights based entirely on financial data stored in PostgreSQL.

2. High-Level Architecture



Flow

1. User submits a question.
 2. Tara Agent identifies the intent.
 3. Appropriate tool is selected.
 4. Tool queries PostgreSQL.
 5. Structured data is returned.
 6. Agent generates a natural language response.
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3. Agent Design

Tara Finance Agent

The Tara Agent acts as the orchestrator.

Responsibilities:

- * Understand user intent
- * Select appropriate tool
- * Convert natural language into structured parameters
- * Generate concise financial insights
- * Format monetary values in INR

Tool Selection

User Intent	Tool
Spending questions	spendingTool
Portfolio questions	portfolioTool
Examples:	
User Query	Tool
How much did I spend on Groceries?	spendingTool
Show my top merchants	spendingTool
What is my largest holding?	portfolioTool
Rank all my funds	portfolioTool

4. Spending Tool Design

The spending tool performs transaction analysis using PostgreSQL.

Supported capabilities:

- * Total spending
- * Category spending
- * Merchant spending
- * Spending trends
- * Spending comparison
- * Category distribution
- * Top merchants
- * Largest transactions

Example Query Flow

User:

Plain text

How much did I spend on Groceries?

Tool Input:

JSON

```
{
  "category": "Groceries",
  "summary": true
}
```

Database Query:

SQL

```
SELECT
COALESCE(SUM(amount),0)
FROM transactions
WHERE category ILIKE '%Groceries%';
```

Result:

JSON

```
{
  "total_spent": 166303.05,
  "category": "Groceries",
  "currency": "INR"
}
```

5. Portfolio Tool Design

The portfolio tool performs investment analysis.

Supported capabilities:

- * Portfolio valuation
- * Largest holding
- * Best performing fund
- * Worst performing fund
- * Fund ranking
- * Holding returns

Performance Calculation

Current Value:

Plain text

$\text{Units} \times \text{Current NAV}$

Purchase Value:

Plain text

$\text{Units} \times \text{Purchase NAV}$

Return Percentage:

Plain text

$$\left(\frac{\text{Current NAV} - \text{Purchase NAV}}{\text{Purchase NAV}} \right) \times 100$$

Profit / Loss:

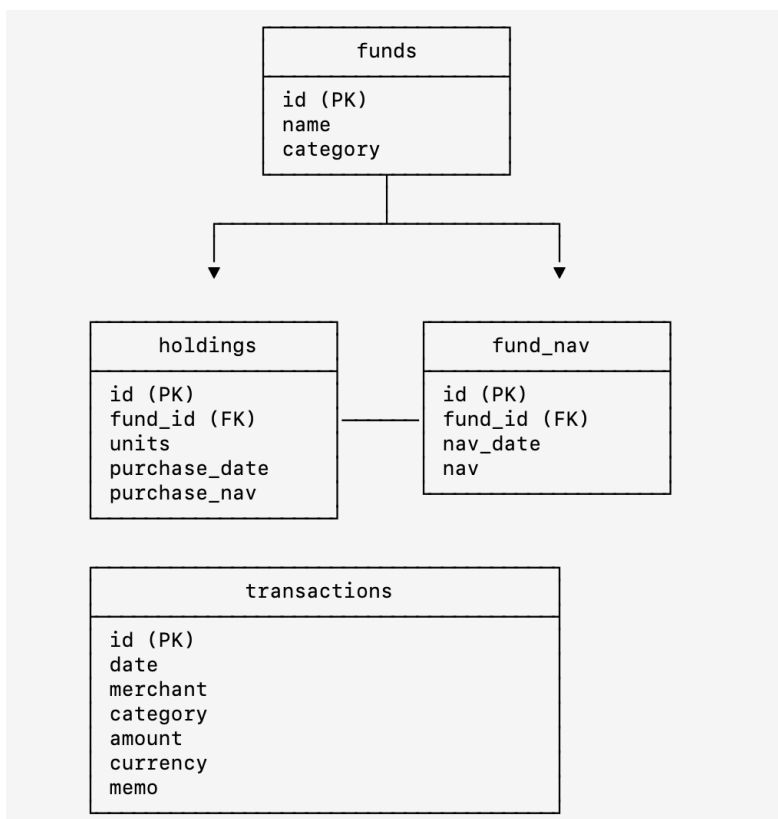
Plain text

$\text{Current Value} - \text{Purchase Value}$

6. Database Design

The solution uses PostgreSQL for financial data storage.

ER Diagram



Relationships

Plain text

```
funds (1) -----< holdings (many)
```

```
funds (1) -----< fund_nav (many)
```

Transactions remain independent because they power spending analytics.

7. Data Ingestion

Financial data is loaded through an ingestion script.

Bash

```
DATA_DIR=./data/sample_a npx tsx scripts/ingest.ts
```

The script loads:

- * transactions.json
- * funds.json
- * holdings.json

into PostgreSQL.

Example result:

Plain text

```
Inserted 1500 transactions
Inserted 8 funds
Inserted 192 NAV records
Inserted 8 holdings
Ingestion complete!
```

8. Design Decisions

PostgreSQL

Chosen because:

- * Strong relational support
- * Efficient aggregation queries
- * Suitable for financial analytics

Tool-Based Architecture

Chosen because:

- * Clear separation of concerns
- * Easier testing
- * Easier future expansion

Mastra Agent

Chosen because:

- * Native tool calling
 - * Structured workflows
 - * Easy integration with TypeScript
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9. Assumptions

- * All monetary values are stored in INR.
 - * Transactions contain valid category and merchant values.
 - * Latest NAV record represents current fund value.
 - * Portfolio calculations are based on available holdings.
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10. Future Improvements

- * Refund detection and adjustment
 - * Transfer exclusion logic
 - * Category growth analytics
 - * Fund performance over custom periods
 - * Multi-user support
 - * Budget tracking and alerts
 - * Portfolio diversification scoring
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Author

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